

PALMERSTON NORTH, New Zealand

WMO: 934040

Lat: 40.323S

Lon: 175.612E

Elev: 128

StdP: 14.63

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.0	34.2	29.3	23.5	34.5	31.2	25.6	36.2	26.3	52.7	23.8	52.9	2.8	110	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.8	78.7	65.7	76.5	64.8	74.3	63.7	68.3	74.9	66.8	73.0	65.4	71.6	11.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.5	71.2	64.4	91.0	70.1	62.8	86.0	69.0	32.7	74.8	31.4	72.9	30.3	71.8	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	22.9	20.5	DB	28.0	83.7	1.3	3.1	27.1	85.9	26.3	87.7	25.6	89.5	24.7	91.7	(4)
(5)			WB	27.4	71.0	1.3	1.6	26.5	72.2	25.7	73.2	25.0	74.1	24.1	75.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	64.1	64.8	61.8	57.0	52.8	48.6	47.8	48.9	51.5	54.0	56.8	61.4	(6)	
	DBStd	7.34	4.63	4.24	4.49	4.75	4.88	4.87	4.44	4.03	4.04	4.06	4.19	4.43	(7)	
	HDD50	331	0	0	0	6	29	84	96	69	30	14	3	0	(8)	
	HDD65	3503	73	50	117	241	377	491	534	500	404	341	248	127	(9)	
	CDD50	2428	436	414	366	216	116	43	27	34	76	139	207	354	(10)	
	CDD65	124	44	44	18	1	0	0	0	0	0	0	1	16	(11)	
	CDH74	529	206	187	70	2	0	0	0	0	0	0	7	57	(12)	
	CDH80	34	18	11	2	0	0	0	0	0	0	0	0	3	(13)	
(14) Wind	WSAvg	9.3	10.0	9.6	9.3	8.2	8.3	8.1	8.4	8.4	9.8	10.4	11.0	9.9	(14)	
(15) Precipitation	PrecAvg	45.1	3.0	3.0	3.0	3.3	3.9	4.4	4.3	4.1	4.1	4.4	3.8	4.0	(15)	
	PrecMax	59.9	5.7	13.5	7.4	9.1	8.5	9.8	7.1	7.6	11.1	9.2	7.2	7.8	(16)	
	PrecMin	35.7	1.2	0.6	1.0	1.1	1.2	2.2	0.9	1.2	0.7	1.7	1.1	1.9	(17)	
	PrecStd	6.0	1.1	2.8	1.4	1.9	1.7	1.6	1.7	1.4	2.0	1.8	1.7	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.3	81.5	79.1	73.2	67.6	62.0	60.8	61.5	65.8	68.9	74.6	78.7	(19)	
		MCWB	67.6	66.4	64.8	63.3	60.2	56.6	55.0	54.7	57.9	59.6	63.9	65.5	(20)	
	2%	DB	78.7	78.6	75.8	69.8	64.6	59.3	58.0	59.3	62.8	65.9	70.5	75.3	(21)	
		MCWB	66.0	65.7	63.7	61.5	59.1	55.1	53.3	53.3	56.2	58.1	61.4	64.0	(22)	
	5%	DB	76.1	76.4	73.3	67.4	62.5	57.6	56.4	57.4	60.8	63.7	67.4	72.8	(23)	
		MCWB	64.8	64.9	62.7	60.0	57.7	54.0	52.2	52.1	55.2	56.9	59.3	63.0	(24)	
	10%	DB	73.5	73.8	70.7	65.2	60.7	56.2	54.7	55.7	58.8	61.5	64.9	70.2	(25)	
		MCWB	63.5	63.7	61.7	59.1	56.6	52.9	50.9	51.3	53.9	55.8	57.9	61.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	70.6	68.0	65.3	62.1	58.1	56.5	56.1	59.4	61.3	65.5	68.3	(27)	
		MCDB	79.4	76.3	74.7	70.1	65.8	59.9	59.4	59.8	63.7	66.5	72.2	74.3	(28)	
	2%	WB	68.4	68.4	65.6	62.9	60.2	56.0	54.0	54.3	57.4	59.2	62.6	66.2	(29)	
		MCDB	75.5	74.6	72.2	67.3	63.3	58.4	57.2	57.8	61.1	63.7	68.6	72.2	(30)	
	5%	WB	66.6	66.8	64.1	61.5	58.6	54.5	52.5	52.9	56.0	57.7	60.7	64.4	(31)	
		MCDB	73.0	72.9	70.7	65.8	61.7	57.1	55.6	56.3	59.7	62.1	65.4	70.2	(32)	
	10%	WB	64.8	65.3	62.8	60.0	56.9	53.1	51.2	51.8	54.6	56.5	59.0	63.0	(33)	
		MCDB	71.3	71.4	69.2	64.2	60.1	55.8	54.1	55.1	58.3	60.8	63.8	68.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	16.8	16.5	15.8	14.0	13.1	12.6	13.3	13.6	13.6	14.4	15.1	(35)	
		MCDBR	20.0	20.4	20.4	18.3	15.2	12.5	12.7	15.0	15.7	16.4	18.0	19.0	(36)	
	5% WB	MCWBR	9.8	9.7	10.2	10.9	10.5	9.5	9.2	10.3	10.5	10.2	10.6	9.8	(37)	
		MCWBR	17.2	15.9	16.5	14.6	12.5	10.6	11.2	12.4	13.5	14.1	15.2	15.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.335	0.318	0.306	0.295	0.291	0.287	0.299	0.324	0.333	0.329	0.345	(40)	
		taud	2.493	2.546	2.588	2.612	2.615	2.600	2.613	2.581	2.487	2.474	2.504	2.492	(41)	
	Ebn at Noon		312	307	299	283	266	255	266	281	292	305	315	314	(42)	
		Edh at Noon	36	32	28	24	20	19	20	24	31	34	35	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2040	1783	1399	953	636	482	561	796	1133	1485	1840	1934	(44)	
	RadStd		136	172	82	67	34	33	43	50	81	93	161	143	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.54	N/A	+0.81	N/A	N/A	N/A	-70	-223	+189	N/A
Regional (8 neighbors)	N/A	+0.58	+0.99	N/A	N/A	N/A	-43	N/A	N/A	N/A

Nomenclature: See separate page